

The Structural Identity: Mass as the Actor and Spacetime as the Subject

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1 The Axiom of Action (Mass vs. Spacetime)

In this structural model, **Mass** is the primary actor (Subject), and **Spacetime** is the affected entity (Object). The identity of curvature is defined by:

$$\rho_m(\text{Actor}) \equiv \mathcal{R}_{str}(\text{Object}) \quad (1)$$

This implies that curvature is not a force field but a structural modification of space caused by density.

2 The Null-Density Theorem (Light)

Since light possesses no rest mass, it possesses zero material density ($\rho_m = 0$). Therefore:

- Light is not an actor; it is the *rate* of spacetime itself.
- Light propagation is a linear projection into the "Object" (Spacetime).

3 Horizon Deception and Density Gradients

The perception of "bending light" is a visual deception (D) caused by the gradient between the high-density "Deep" and the low-density "Horizon" (Earth).

$$D_{star} = \arctan(n_{depth}) \cdot \cos(\phi) \cdot \frac{\rho_{Actor}}{\rho_{Observer}} \quad (2)$$

4 Conclusion: The Fate of Information

At the Event Horizon, the Actor (Mass) and the Light (Rate) collide, leading to **Electron Vaporization**. The signal intensity drops to zero ($SI \rightarrow 0$), not by capture, but by phase transformation into the null-state, allowing for Galactic Expansion.